# IBM University Engagement Project Submission

*"Exploring the Power of Agentic AI with IBM Granite and LangFlow"*

**Project Tile – Smart Farming Advice Agent**

**Domain of  Project – Agriculture**

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# Problem Statement -

## Problem Statement: Smart Farming Advice Agent

### The Challenge

**Farmers often face difficulties in identifying crop diseases, selecting suitable fertilizers, managing irrigation, and obtaining timely agricultural guidance. Access to expert advice is limited, especially in rural areas, resulting in reduced crop productivity and inefficient resource utilization.**

### The Objective

**Develop an AI-powered Smart Farming Advisor that provides farmers with crop management recommendations, pest and disease guidance, irrigation suggestions, and fertilizer recommendations. The system should deliver simple, practical, and timely advice to support informed agricultural decision-making.**

# Technology Used

- LangFlow – Visual workflow design and orchestration platform
- IBM WatsonX (Planned Integration) – Platform for accessing IBM Granite models
- IBM Granite Models (Planned Integration) – Large Language Models for agricultural recommendations and reasoning
- Agentic AI – Autonomous decision-making and recommendation generation
- Prompt Engineering – Structured agricultural advisory generation

# Proposed Solution -

**The proposed solution is an AI-powered Smart Farming Advisor developed using LangFlow and designed for integration with IBM Granite models through IBM WatsonX.**

**The system accepts farming-related queries and provides recommendations regarding crop management, pest control, disease identification, irrigation planning, and fertilizer usage. A prompt-driven agent processes user inputs and generates practical suggestions that farmers can easily understand and implement.**

**The solution uses a modular workflow consisting of Chat Input, Agent, Prompt Template, and Chat Output components. This architecture enables future integration with external agricultural databases, weather information services, and IBM Granite models for enhanced intelligence and scalability.**

**The system aims to support farmers in making informed agricultural decisions and improving productivity through AI-assisted guidance.**

# Langflow component Used -

1. Chat Input – Accepts farming-related questions from users.

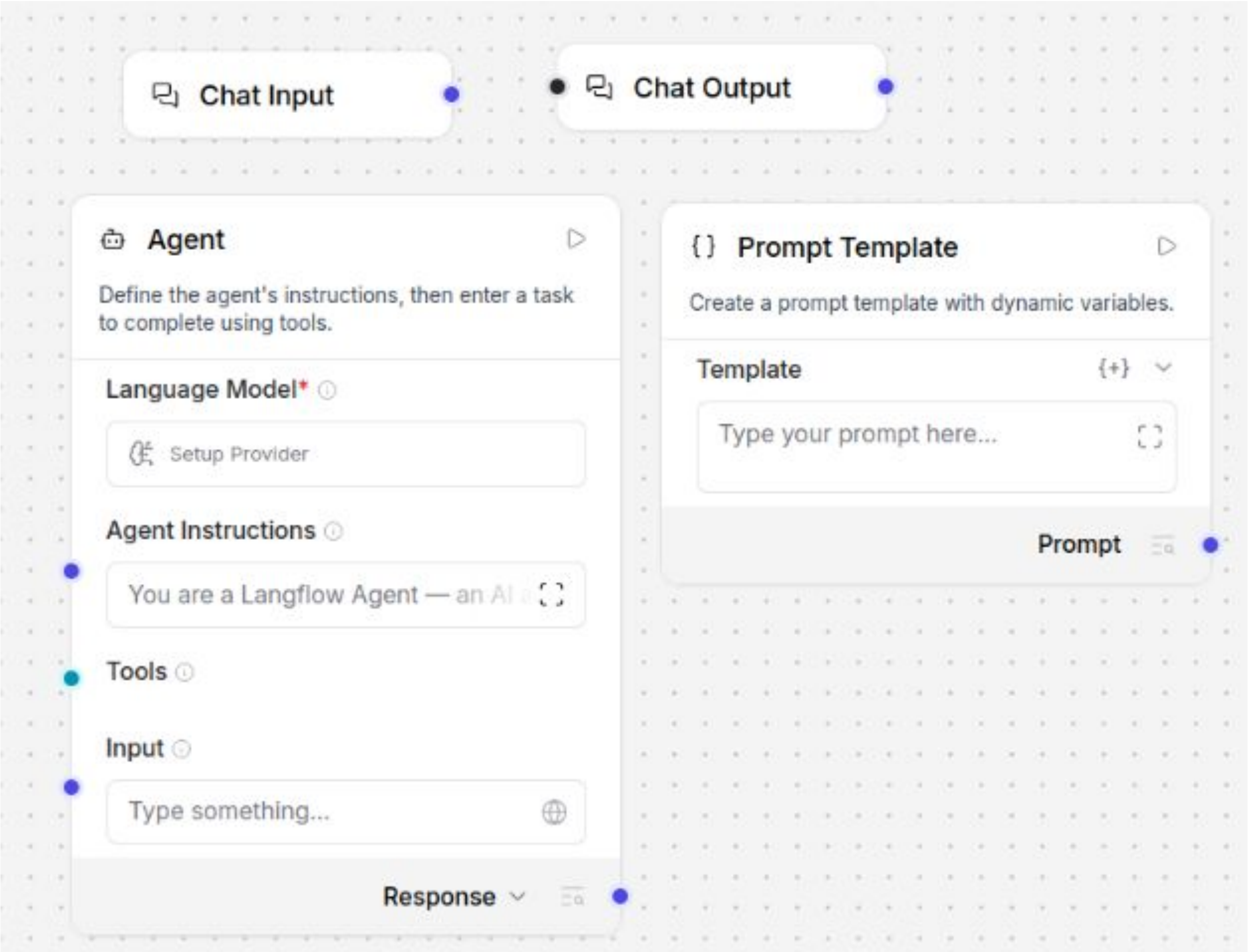
2. Agent – Processes user requests and generates agricultural recommendations.

3. Prompt Template – Defines farming-specific instructions and response behavior.

4. Chat Output – Displays generated recommendations.

5. IBM WatsonX Provider – Intended integration point for IBM Granite models.

# Langflow workflow -



## Architecture blueprint

Farmer/User

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Chat Input

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LangFlow Agent

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Prompt Template

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IBM Granite Model

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Agricultural Recommendation

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Chat Output

# Role of Agentic AI in the solution

**Role of Agentic AI in Smart Farming Advisor**

**Agentic AI enables the Smart Farming Advisor to function as an intelligent decision-support system rather than a simple chatbot. The agent receives farming-related queries, interprets user requirements, and generates context-aware recommendations.**

**The system can assist farmers with crop management, pest and disease identification, irrigation planning, and fertilizer recommendations. By using structured prompts and agent-based reasoning, the solution provides practical and easy-to-understand agricultural guidance.**

**Agentic AI allows the system to automate information processing, improve recommendation quality, and support informed agricultural decision-making. Future integration with IBM Granite models and agricultural knowledge sources can further enhance the intelligence and adaptability of the system.**

# Technology Used

• LangFlow – Visual platform for designing and orchestrating AI workflows.

• IBM WatsonX – Intended platform for deploying and accessing IBM Granite models.

• IBM Granite Models – Large Language Models used for reasoning and agricultural recommendation generation.

• Agentic AI – Enables autonomous task execution and recommendation generation.

• Prompt Engineering – Provides domain-specific farming instructions to the AI agent.

• Python Environment – Used to deploy and run LangFlow locally.

# Token Usage

Prototype Demonstration

• Local LangFlow workflow designed and tested.

• No external API consumption during prototype development.

• Token usage will depend on the selected IBM Granite model during deployment through IBM WatsonX.

# Novelty and Uniqueness

• Agriculture-Focused Advisory System – Designed specifically for farming assistance.

• Agent-Based Architecture – Uses an AI agent to process and respond to agricultural queries.

• Modular LangFlow Workflow – Components can be extended with additional tools and knowledge sources.

• IBM Granite Ready – Designed for future integration with IBM Granite models through WatsonX.

• User-Friendly Guidance – Generates practical recommendations understandable by farmers.

• Scalable Design – Can be extended with weather data, soil information, and crop databases.

# Git Hub Link

https://github.com/niyatibuch/smart-farming-advisor-langflow

# Future Scope

1. Weather Data Integration

The system can integrate real-time weather information to provide more accurate irrigation and crop management recommendations.

2. Crop Disease Detection

Image-based disease identification can be incorporated using computer vision models to assist farmers in diagnosing crop problems.

3. Regional Language Support

Support for multiple regional languages can improve accessibility for farmers across different regions.

4. IBM Granite Integration

Future deployment with IBM Granite models through WatsonX will enhance reasoning capability and recommendation quality.

5. Agricultural Knowledge Base

Integration with agricultural databases and government advisory systems can improve recommendation accuracy.

# Thank You!

Thank you for your time and interest.